Molecules and Cells

Minireview

Human γδ T cells in the tumor microenvironment: Key insights for advancing cancer immunotherapy

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ABSTRACT

The role of γδ T cells in antitumor responses has gained significant attention due to their major histocompatibility complex (MHC)-independent killing mechanisms, which are functionally distinct from conventional αβ T cells. Notably, γδ tumor-infiltrating lymphocytes (TILs) have been identified as favorable prognostic markers in various cancers. However, the γδ TIL subsets, including Vδ1, Vδ2, and Vδ3, exhibit distinct prognostic implications and phenotypes within the tumor microenvironment (TME). Although the underlying mechanisms remain unclear, recent studies suggest that these subset-specific differences may arise from divergent activation pathways. Vδ1 TILs appear to be mainly activated by γδ T-cell receptor (TCR) signaling, whereas Vδ2 TILs seem to rely on alternative pathways, such as natural killer (NK) receptor-mediated activation. In addition to phenotypic studies, cancer immunotherapies, such as engineered γδ T cells, γδ T-cell engagers, and γδ TCR–based therapies, are under active development. However, despite these advancements, functional heterogeneity and limited persistence within TME remain significant challenges. Overcoming these obstacles could position γδ T-cell therapies as a transformative platform for cancer treatment. Here, we review recent findings on the prognostic significance of human γδ T cells, their phenotypic characteristics, and advances in γδ T-cell therapies, offering valuable insights for the development of novel cancer immunotherapies.

Keywords: γδ tumor-infiltrating lymphocytes, Advancements in γδ T-cell therapy, Limitations of γδ T-cell therapy, Phenotypic characteristics of γδ T cells, Prognostic value of γδ T cells

INTRODUCTION

γδ T cells are an unconventional subset of T lymphocytes characterized by the expression of a γδ T-cell receptor (TCR), which distinguishes them from conventional αβ T cells (Hayday et al., 1985). In humans, γδ T cells are classified based on TCRδ chains, with Vδ1, Vδ2, and Vδ3 being the most frequently utilized among the 8 known Vδ variants. Vδ2 T cells constitute 60% to 95% of the circulating γδ T-cell population in adults and predominantly pair with the Vγ9 chain, resulting in Vγ9Vδ2 T cells. These cells are mainly activated by unconventional antigens, such as the butyrophilin subfamily 3 member A (BTN3A) (Mensurado et al., 2023). Although the 3 BTN3A isoforms (BTN3A1, BTN3A2, and BTN3A3) share highly similar ectodomains, only BTN3A1 contains a high-affinity phosphoantigen (pAg)-binding B30.2 domain within its cytoplasmic region. This unique domain enables BTN3A1 to sense an increase in intracellular pAg (Sandstrom et al., 2014). Within the tumor microenvironment (TME), dysregulated mevalonate pathways within the tumor cells lead to pAg accumulation, inducing conformational change in BTN3A1 ectodomains (Gober et al., 2003). This alteration enables BTN3A1 to form a complex with BTN2A1, which directly activates the Vγ9Vδ2 TCR (Rigau et al., 2020).

In contrast, Vδ1 T cells comprise up to one-third of circulating γδ T cells, and are preferentially located in peripheral tissues, particularly within epithelial and mucosal tissues (Mensurado et al., 2023). Unlike the semi-invariant Vδ2 TCR, Vδ1 TCRs exhibit considerable clonal diversity and can recognize lipid-presenting molecules such as CD1c and CD1d (Adrienne et al., 2013; Russano et al., 2007).

Vδ3 T cells, while rare in peripheral blood under homeostatic conditions, exhibit a strong liver tropism (Kenna et al., 2004). Vδ3 TCRs can recognize antigens such as Annexin A2 and the major histocompatibility complex (MHC) class 1–related protein MR1 (Marlin et al., 2017; Rice et al., 2021). Moreover, although Vδ3 T cells can recognize CD1d similarly to Vδ1 T cells, they do not interact with CD1c (Mangan et al., 2013). However, the antigens recognized by Vδ1 or Vδ3 TCRs within the TME remain unclear.

Although γδ T-cell subsets are not conserved between humans and mice, these cells share functional characteristics across species, including tissue residency, proinflammatory cytokine production, and cytotoxic activity. Consequently, γδ T cells contribute to diverse immune responses against stress, infections, and malignancies in both humans and mice (Park and Lee, 2021). In particular, the antitumor roles of γδ T cells have garnered significant attention due to their unique features, such as MHC-independent killing mechanisms and diverse receptor-ligand interactions (De Vries et al., 2023; Silva-Santos et al., 2019). These characteristics enable γδ T cells to be developed as allogeneic therapies targeting various tumors (Neelapu et al., 2022; Xu et al., 2021). Moreover, their propensity for peripheral migration makes γδ T cells suitable for solid tumor therapy, as they can recognize antigens directly within the TME (Willcox et al., 2020). Thus, γδ T cells are being explored as promising agents in cancer immunotherapy. In this review, we discuss recent findings from a comprehensive analysis of γδ T cells within the human TME and highlight advanced strategies to enhance γδ T-cell therapy.

HUMAN ΓΔ TUMOR-INFILTRATING LYMPHOCYTES AS PROGNOSTIC MARKERS IN TUMORS

A comprehensive pan-cancer analysis utilizing the CIBERSORT algorithm identified γδ T cells as the most favorable prognostic factor across various tumors in a cohort of 18,000 patients (Gentles et al., 2015). Although the CIBERSORT algorithm has limitations in accurately distinguishing γδ T cells from CD8 T cells, recent studies have confirmed that higher levels of γδ tumor-infiltrating lymphocytes (TILs) are positively associated with better prognosis in various tumors, including colorectal cancer (CRC), bladder cancer, glioma, and hepatocellular carcinoma (HCC) (Meraviglia et al., 2017; Nguyen et al., 2022; Park et al., 2021; Zakeri et al., 2022).

Initially, Vδ2 T cells garnered interest due to their high abundance in peripheral blood and ease of accessibility compared with other γδ T-cell subsets (Bonneville and Scotet, 2006). However, recent research has shifted attention toward Vδ1 T cells, which have emerged as significant prognostic markers in multiple cancers, even in cases where Vδ2 T cells showed no prognostic value. High frequencies of Vδ1 TILs have been associated with improved overall survival (OS) in patients with certain cancers, such as epithelial ovarian cancer (EOC), endometrial cancer, lung adenocarcinoma, non-small-cell lung cancer (NSCLC), and triple-negative breast cancer (Foord et al., 2021; Harmon et al., 2023; Wu et al., 2019, 2022). Furthermore, the Vδ2$^-$ T-cell signature score, derived from a comparative analysis of Vδ2$^-$ T cells and CD8 T cells, has been identified as a favorable prognostic marker in renal cell carcinoma (RCC) (Rancan et al., 2023). Because the Vδ2$^-$ T-cell population predominantly consists of Vδ1 T cells, these studies also underscore the prognostic value of Vδ1 TILs. Additionally, a high frequency of CD69$^+$ Vδ1 effector TILs, but not total T cells, has been positively correlated with OS in patients with colon liver metastasis (CLM) (Bruni et al., 2022). These findings suggest that identifying functional subsets of γδ TILs could enhance the precision of γδ T-cell biomarkers in cancer prognosis.

In contrast to Vδ1 TILs, the frequency of Vδ2 TILs does not correlate with OS in most cancers, likely due to their low abundance among γδ TILs. However, a recent study demonstrated a strong positive association between the antitumor cytokine production of Vδ2 TILs after in vitro stimulation and OS in patients with EOC (Foord et al., 2021). This suggests that the functional capability of the Vδ2 TILs may be more important than their quantity.

Interestingly, γδ TILs have also been identified as prognostic indicators for antitumor therapy responses. High Vδ1 TIL frequency has been linked to favorable responses to anti-programmed cell death protein 1 (PD-1) therapy in melanoma and advanced solid cancers (Davies et al., 2024; Wu et al., 2022). Similarly, the Vδ2$^-$ T-cell signature score was higher in anti-PD-1 responders with RCC (Rancan et al., 2023). In contrast, T-cell immunoreceptor with Ig and ITIM domains (TIGIT)$^+$ γδ T cells, which exhibit an exhaustion phenotype akin to conventional αβ T cells, showed a negative correlation with chemotherapy responses in patients with acute myeloid leukemia (AML) (Hou et al., 2024).

Together, these findings highlight γδ T cells as crucial prognostic markers for both survival outcomes and antitumor therapy responses in cancer patients, underscoring their significant roles in tumor immunology.

PHENOTYPIC CHARACTERISTICS OF HUMAN ΓΔ T CELLS IN TUMOR MICROENVIRONMENT

Vδ1 T Cells

Vδ1 and Vδ2 T cells share several functional similarities, including antitumor cytokine production and cytotoxic activity. However, recent studies have highlighted distinct phenotypic characteristics between Vδ1 and Vδ2 TILs (Table 1). In EOC, a negative correlation between TCR clonality and activation marker expression was observed in Vδ1 TILs, whereas this relationship was not evident in Vδ2 TILs. These findings suggest that the activation of Vδ1 TILs, but not Vδ2 TILs, is predominantly driven by TCR stimulation (Foord et al., 2021). Additionally, research on patients with Merkel cell carcinoma (MCC) revealed that the 4 predominant TCR clonotypes, comprising 69% of the total γδ TIL population, are Vδ1 TCR clones, indicating TCR-driven clonal expansion (Gherardin et al., 2021). Further evidence of TCR-driven activation of Vδ1 TILs comes from studies showing that their activation states resemble those of conventional αβ TILs. For instance, a substantial proportion of Vδ1 TILs in HCC or CLM were found to differentiate into effector memory T cells re-expressing CD45RA (T_{EMRA}), representing terminally differentiated effector cells (Bruni et al., 2022; Zakeri et al., 2022).

Moreover, Vδ1 TILs express exhaustion markers typical of conventional αβ TILs, such as PD-1, T-cell immunoglobulin, and mucin domain 3 (TIM-3), and TIGIT across various tumors. Notably, several studies have demonstrated the response of Vδ1 TILs to anti-PD-1 therapy in melanoma, microsatellite instability CRC, MCC, and RCC (Davies et al., 2024; De Vries et al., 2023; Lien et al., 2024; Rancan et al., 2023). These

Table 1. Comparison of phenotypic characteristics between Vδ1 and Vδ2 TILs

Phenotype	Vδ1 TILs	Vδ2 TILs	Cancer	References
TCR clonality and activation state	Negative correlation	No correlation	EOC	Foord et al. (2021)
Effector state	Predominantly T_{EMRA} ($CD45RA^{+}$, $CD27^{-}$)	Limited T_{EMRA} Mostly T_{EM} ($CD45RA^{-}$, $CD27^{-}$)	CLM MSS CRC HCC	Bruni et al. (2022) Stary et al. (2024) Zakeri et al. (2022)
Exhaustion markers	High expression (PD-1, TIM-3, etc.)	Low expression	MSI CRC EOC RCC	De Vries et al. (2023) Foord et al. (2021) Rancan et al. (2023)
Response to anti-PD-1 therapy	Responsive	No observed response	Melanoma MSI CRC RCC	Davies et al. (2024) De Vries et al. (2023) Rancan et al. (2023)
Proliferative capacity	Expression of Ki67 TCR-driven clonal expansion	Absence of Ki67	MSI CRC MCC MSS CRC	De Vries et al. (2023) Gherardin et al. (2021) Stary et al. (2024)
Frequency in tumor tissues	Most abundant γδ TIL subset	Relatively low frequency	CLM CRC MSS CRC	Bruni et al. (2022) Harmon et al. (2023) Stary et al. (2024)
Functional heterogeneity	Consist of both antitumor ($IFN\text{-}\gamma^{+}$) and protumor ($AREG^{+}$) subsets	Mainly antitumor subsets	CRC	Harmon et al. (2023)

TILs, tumor-infiltrating lymphocytes; TCR, T-cell receptor; T_{EMRA}, effector memory T cells re-expressing CD45RA; T_{EM}, effector memory T cells; EOC, epithelial ovarian cancer; CLM, colon liver metastasis; MSS, microsatellite stable; CRC, colorectal cancer; HCC, hepatocellular carcinoma; MSI, microsatellite instability; RCC, renal cell carcinoma; MCC, Merkel cell carcinoma; IFN-γ, interferon-γ; AREG, amphiregulin.

observations imply that Vδ1 TILs are functionally impaired, and reversing this impairment could significantly enhance their antitumor activity, leading to improved immunotherapy outcomes.

Vδ2 T Cells

In contrast to Vδ1 TILs, Vδ2 TILs exhibit distinctive characteristics compared with conventional αβ TILs. Notably, Vδ2 TILs predominantly differentiate into effector memory T cells (T_{EM}), with a significantly reduced population of T_{EMRA} cells compared to Vδ1 TILs in patients with HCC or CLM (Bruni et al., 2022; Zakeri et al., 2022). These findings suggest that Vδ2 TILs reside in a less terminally differentiated effector state.

Furthermore, Vδ2 TILs express low levels of exhaustion markers in various cancers. Although low levels of exhaustion markers may indicate a lower activation status, Vδ2 TILs demonstrate cytotoxicity and antitumor cytokine production comparable to the levels in Vδ1 TILs (Foord et al., 2021; Rancan et al., 2023). These observations indicate that the activation status of Vδ2 TILs is sufficient to support an effective antitumor immune response, even without evidence of chronic TCR stimulation. Moreover, a recent study highlighted the importance of alternative activation pathways, such as natural killer (NK) receptors like CD226 (DNAM-1), in their antitumor function (Choi et al., 2022).

However, a significant challenge for Vδ2 T cells is their limited proliferative capacity, as evidenced by the absence of Ki67 expression in Vδ2 TILs in patients with CRC (De Vries et al., 2023; Stary et al., 2024). Additionally, the frequency of Vδ2 T cells within the TME was markedly lower compared with adjacent nontumor tissues or healthy individuals (Bruni et al., 2022; Harmon et al., 2023). This limited proliferative potential may hinder Vδ2 TILs from sustaining robust antitumor responses within the highly suppressive TME. Therefore, Vδ2 T cells hold promise as candidates for cancer immunotherapy, particularly when combined with strategies designed to enhance their persistence within the TME.

Functional Heterogeneity of Human Vδ1 and Vδ2 T Cells

In murine models, the primary mechanism by which γδ T cells exhibit protumor effects is through interleukin-17 (IL-17) production (Park and Lee, 2021). Although IL-17 production by human γδ TILs has been previously reported in CRC (Wu et al., 2014), recent studies have shown that IL-17A transcripts were undetectable in Vδ1 or Vδ2 TILs within CRC (Harmon et al., 2023). Furthermore, both $V\delta2^{+}$ and $V\delta2^{-}$ TILs in RCC displayed minimal IL-17 production compared with conventional αβ TILs (Rancan et al., 2023). Similarly, Vδ1 and Vδ2 TILs failed to produce IL-17 following in vitro stimulation (Cazzetta et al., 2021; Wu et al., 2022). Thus, although some controversy remains regarding the IL-17 production by Vδ2 TILs, current evidence indicates either no or minimal protumor activity from these cells.

However, recent findings suggest that Vδ1 TILs can differentiate into protumor subsets depending on the tumor subtypes. In endometrial cancer, where Vδ1 TILs are associated with favorable prognostic outcomes, these cells exclusively consisted of interferon-γ $(IFN\text{-}\gamma)^{+}$ antitumor populations. In contrast, Vδ1 TILs in CRC were heterogeneous, consisting of both $IFN\text{-}\gamma^{+}$ antitumor and amphiregulin $(AREG)^{+}$ protumor subsets. Notably, the frequency of Vδ1 TILs did not correlate with OS in patients with CRC (Harmon et al., 2023). These findings indicate that the local TME plays a critical role in shaping the differentiation of Vδ1 TILs, and influencing their overall immune response to tumors. Given the diverse stimuli impacting Vδ1 TILs beyond TCR stimulation (Mensurado et al., 2024), identifying the key factors that drive Vδ1 TILs toward either a protumor or an antitumor phenotype is essential for unlocking their full immunotherapeutic potential.

Table 2. Summary of completed and ongoing clinical trials for γδ T-cell therapy in cancer

Type of therapy	Start year	Phase	Status	Disease(s)	Patients	Outcome	Clinical trial ID (References)
In vivo stimulation							
Zoledronate + IL-2 + Ca/vitamin D	2007	Phase I	Completed	Prostate cancer	9	4 SD, 2 PR	(Dieli et al., 2007)
Zoledronate + IL-2	2005	Phase I	Completed	Breast cancer	5	2 SD, 1 PR	(Meraviglia et al., 2010)
	2011	Phase I	Completed	Neuroblastoma	4	1 SD	NCT01404702 (Pressey et al., 2016)
	2012	Phase I/II	Completed	RCC	7	3 SD	(Kunzmann et al., 2012)
				Melanoma	6	1 SD	
				AML	8	2 SD, 2 PR	
Adoptive γδ T-cell therapy							
Autologous Vγ9Vδ2 T cells	2006	Phase I	Completed	NSCLC	12	6 SD	(Sakamoto et al., 2011)
	2012	Phase II	Completed	NSCLC	25	16 SD, 1 PR	UMIN000006128 (Kakimi et al., 2020)
Autologous Vγ9Vδ2 T cells + zoledronate	2011	Phase I	Completed	Melanoma	6	2 SD	(Nicol et al., 2011)
Allogeneic Vγ9Vδ2 T cells + locoregional therapy	2017	Phase I/II	Completed	Lung cancer	9	1 SD; median OS: 19.1 vs 9.1 months (control)	NCT03183232 (Xu et al., 2021)
	2017	Phase I/II	Completed	ICC	14	Median PFS: 8 vs 4 months (control)	NCT03183219 (Zhang et al., 2022)
				HCC	15	Median PFS: 8 vs 4 months (control); median OS: 13 vs 8 months (control)	
Allogeneic chemotherapy-resistant γδ T cells + HSCT + chemotherapy	2020	Phase I	Ongoing	Hematological malignancies	10	10 CR	NCT03533816 (INB-100) (McGuirk et al., 2024)
Allogeneic α-CD20 CAR Vδ1 T cells	2021	Phase I	Ongoing	B-cell malignancies	6	4 CR	NCT04735471 (ADI-001) (Neelapu et al., 2022)
Allogeneic Vδ1 T cells	2023	Phase I/IIa	Ongoing	AML	–	–	NCT05886491 (GDX012)
Allogeneic α-B7-H3 CAR γδ T cells	2023	Phase I/IIa	Ongoing	B7-H3$^+$ recurrent GBM	7	3 ORR	NCT06018363 (QH104) (Li et al., 2024)
Allogeneic α-PD-1-secreting pan-γδ T cells	2024	Phase I	Ongoing	Solid tumors	–	–	NCT06404281 (γδ T-PD-1 Ab cells) (Wang et al., 2023)
Allogeneic α-HLA-G CAR BiTE-secreting Vγ9Vδ2 T cells	2024	Phase I/IIa	Ongoing	NSCLC, TNBC, CRC, and GBM	–	–	NCT06150885 (CAR001) (Huang et al., 2023)
Allogeneic NKG2DL-targeting CAR Vγ9Vδ2 T cells	2024	Phase I	Ongoing	Solid tumors, hematologic malignancies	–	–	NCT05302037 (CTM-N2D)
γδ T-cell engager							
Vδ2 TCR × CD1d bispecific antibody	2021	Phase I/IIa	Ongoing	r/r CLL, MM, and ALL	–	–	NCT04887259 (LAVA-051) (De Weerdt et al., 2021)
Vδ2 TCR × EGFR bispecific antibody	2023	Phase I	Ongoing	CRC, NSCLC, HNSCC, and PDAC	–	–	NCT05983133 (PF08046052) (King et al., 2023)
Vδ2 TCR × CD123 bispecific antibody	2024	Phase I	Ongoing	CD123$^+$ AML, MDS	–	–	ACTRN12624001214527 (LAVA-1266)

(*continued on next page*)

Table 2 (*continued*)

Type of therapy	Start year	Phase	Status	Disease(s)	Patients	Outcome	Clinical trial ID (References)
Vδ2 TCR × CD33 bispecific antibody	2025	Phase I	Ongoing	CD33$^+$ AML	–	–	NCT06618001 (JNJ-89853413)
γδ TCR-based therapy							
Conventional αβ T cells with defined Vγ9Vδ2 TCR	2017	Phase I	Ongoing	AML, MDS, and MM	6	2 SD, 1 CR	NL6357 (TEG001) (Straetemans et al., 2018)

CR, complete response; PR, partial response; SD, stable disease; OS, overall survival; PFS, progression-free survival; RCC, renal cell carcinoma; AML, acute myeloid leukemia; NSCLC, non-small-cell lung cancer; ICC, intrahepatic cholangiocarcinoma; HCC, hepatocellular carcinoma; HSCT, hematopoietic stem cell transplantation; GBM, glioblastoma; TNBC, triple-negative breast cancer; CRC, colorectal cancer; r/r CLL, relapsed/refractory chronic lymphocytic leukemia; MM, multiple myeloma; ALL, acute lymphoblastic leukemia; HSNCC, head and neck squamous cell carcinoma; PDAC, pancreatic ductal adenocarcinoma; MDS, myelodysplastic syndrome; MM, multiple myeloma; EGFR, epidermal growth factor receptor.

Vδ3 T Cells

Vδ3 TILs represent the smallest population among γδ TILs compared with Vδ1 and Vδ2 TILs in most cancers (De Vries et al., 2023; Rancan et al., 2023). However, they exhibit a higher frequency than Vδ2 TILs in liver cancer, suggesting that the frequency of Vδ3 TILs may vary depending on the tissue due to tissue tropism (Bruni et al., 2022).

In some studies, Vδ3 TILs show similar phenotypes to Vδ1 TILs, including tumor killing, response to anti-PD-1 therapy, and the presence of AREG$^+$ protumor subsets within CRC (De Vries et al., 2023; Harmon et al., 2023). However, unlike Vδ1 TILs, none of the recent studies demonstrated the prognostic significance of Vδ3 TILs, indicating that their role in the TME may differ from Vδ1 TILs.

Interestingly, a study on CLM patients showed unique features of Vδ3 TILs such as high expression of MHC class II–related genes and heat shock protein–related genes (Bruni et al., 2022). However, the mechanisms and roles of Vδ3 TILs remain unresolved due to a lack of data. Furthermore, the low frequency of Vδ3 T cells in healthy peripheral blood poses challenges in their accessibility for therapeutic studies. Consequently, Vδ3 TILs are not further considered in subsequent sections of this review.

STRATEGIES FOR USING ΓΔ T CELLS AS CANCER IMMUNOTHERAPY

γδ T cells are emerging as promising immunotherapeutic agents due to their allogeneic compatibility, ability to recognize a broad range of antigens, and capacity to target diverse tumor cells.

Traditional therapeutic approaches using Vγ9Vδ2 T cells often involve in vivo stimulation with amino-bisphosphonates. Among these agents, zoledronate has been identified as the most effective drug for interfering with mevalonate pathway, thereby inducing BTN3A1 conformational changes that are recognized by the Vγ9Vδ2 TCR (Nicol et al., 2011). Furthermore, zoledronate can be used to selectively expand Vγ9Vδ2 T cells ex vivo for adoptive cell therapy, and it can be combined with adoptive Vγ9Vδ2 T-cell therapy to enhance therapeutic efficacy in vivo (Wada et al., 2014).

By contrast, Vδ1 T cells, which have recently gained attention in immunotherapy, are also being developed for adoptive cell therapy. Unlike Vγ9Vδ2 T cells, a specific chemical or drug targeting the Vδ1 TCR has not yet been identified. Therefore, current Vδ1 T-cell therapies undergoing clinical trials use anti–CD3 antibody following depletion of conventional αβ T cells or anti–Vδ1 antibody for ex vivo expansion. Additionally, cytokine combinations such as IL-1β, IL-4, IL-15, IL-21, and IFN-γ are utilized for optimal ex vivo expansion conditions (Nishimoto et al., 2022; Sánchez Martínez et al., 2022).

Given that γδ T cells have emerged as key immune cells in the antitumor response within the TME, there has been an increasing number of clinical trials investigating these therapies. A summary of completed and ongoing clinical trials for γδ T-cell therapy in cancer is provided in Table 2. However, traditional strategies have so far demonstrated limited objective responses, likely due to the immunosuppressive TME, excessive polyclonality of the Vγ9Vδ2 TCR, and short persistence of γδ T cells (Park and Lee, 2021). To overcome these challenges and

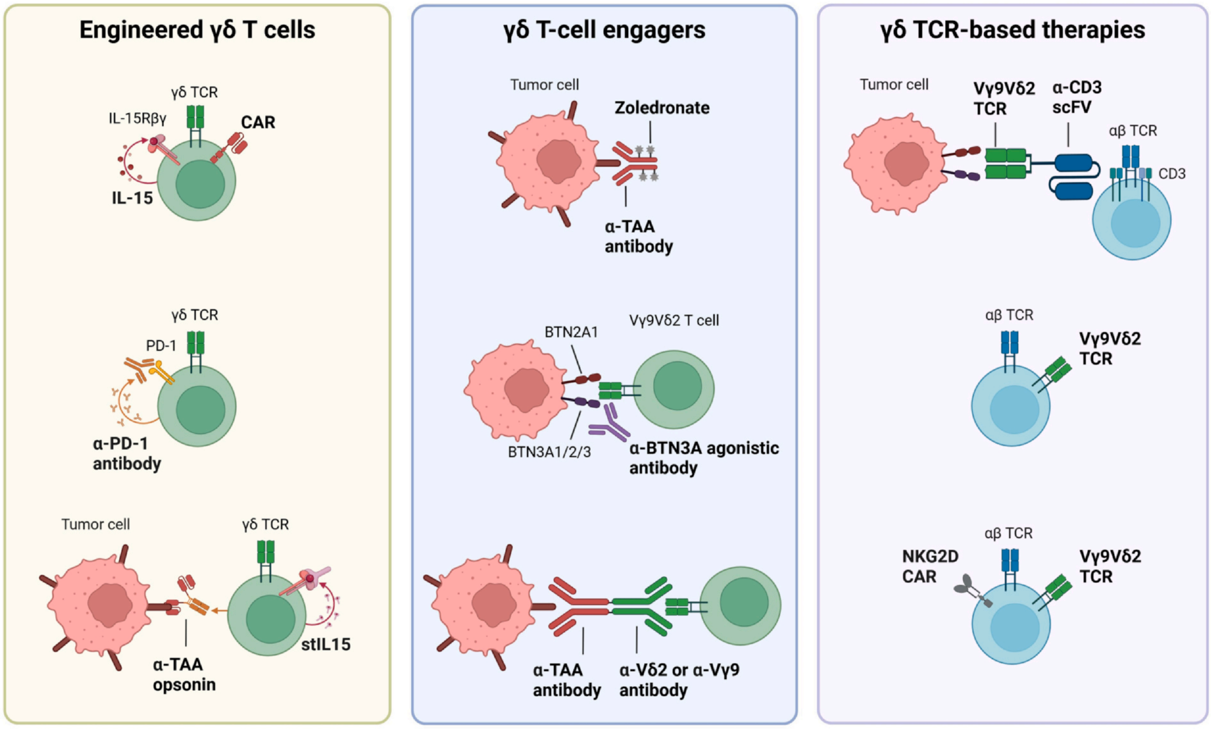

Fig. 1. Recent advances in γδ T-cell therapies. **Engineered γδ T cells:** γδ T cells, known for their allogeneic compatibility, are utilized in adoptive cell therapy, where chimeric antigen receptors (CARs) can be introduced to target tumor-associated antigens (TAAs). These cells can be further engineered to overexpress IL-15, in order to enhance in vivo persistence. Other strategies involve anti-PD-1 antibody–secreting γδ T cells to counteract immunosuppressive tumor microenvironment. Additionally, γδ T cells can be engineered to secrete anti-TAA opsonins, which are fusion proteins combining TAA-targeting single-chain variable fragment (scFv) and the Fc portion of the antibody. These engineered cells also produce stabilized IL-15 (stIL15), a fusion protein of IL-15 with the sushi domain of IL-15Rα, to improve cell persistence. **γδ T-cell engagers:** γδ T cells can be activated in vivo using various γδ T-cell engagers. Zoledronate, a traditional engager, induces a conformational change in BTN3A1 specifically on tumor cells when conjugated with an anti-TAA antibody. Additionally, anti-BTN3A agonistic antibodies can induce conformational changes in all BTN3A isoforms recognized by Vγ9Vδ2 T cells. Other engager molecules include bispecific antibodies targeting both TAAs and Vγ9Vδ2 TCR, which enhance the tumor-targeting capabilities of Vγ9Vδ2 T cells. **γδ TCR-based therapies:** Alternative approaches involve equipping conventional αβ T cells with the Vγ9Vδ2 TCR to enhance their tumor-targeting abilities. This can be achieved using γδ TCR-antibody bispecific molecules (GABs), which consist of the Vγ9Vδ2 TCR and an anti-CD3 scFv, or by engineering autologous αβ T cells to express a defined Vγ9Vδ2 TCR (TEGs). TEGs can be further engineered to express NKG2D CAR, thereby enhancing their antitumor responses.

enhance the therapeutic potential of γδ T cells, novel strategies are being developed and are currently undergoing clinical trials.

In the following sections, we discuss recent advances in engineered γδ T cells, γδ T-cell engagers, and γδ TCR-based therapies as outlined in Figure 1.

Engineered γδ T Cells

First, to enhance the cytolytic effects of γδ T cells, recent studies have developed chimeric antigen receptor (CAR) Vδ1 T cells targeting glypican-3 (GPC-3) for HCC or CD123 for the treatment of AML (Makkouk et al., 2021; Sánchez Martínez et al., 2022). Additionally, ADI-001, an engineered therapy consisting of allogeneic α-CD20 CAR Vδ1 T cells, has demonstrated in vivo antitumor responses in a B-cell lymphoma xenograft mouse model. Consequently, ADI-001 is currently undergoing clinical trial for safety and efficacy (NCT04735471) (Nishimoto et al., 2022).

Another study developed engineered Vγ9Vδ2 T cells capable of secreting humanized anti-PD-1 antibodies to overcome the immunosuppressive effects within the TME. These cells, referred to as Lv-PD-1-γδ T cells, enhanced their activation by blocking immune checkpoints (Wang et al., 2023). Furthermore, a recent study developed disialoganglioside (GD2)-specific opsonin-secreting Vγ9Vδ2 T cells. These cells exhibited not only enhanced cytotoxicity but also the potential to activate bystander immune cells, such as NK cells (Fowler et al., 2024).

However, all of these newly developed cells face the challenge of limited persistence within the TME. To address this limitation, some studies have explored the use of cytokines as combinatorial therapies or engineered γδ T cells to overexpress cytokines, such as IL-15. These strategies have shown promise in immunodeficient NOD.Cg-*Prkdc*^{scid} *Il2rg*^{tm1Wjl}/SzJ (NSG) mice, which are commonly used for patient-derived xenograft models. Nevertheless, administered or overexpressed cytokines can be inefficient within immunocompetent TME due to competition among various immune cells (Allen et al., 2022). Consequently, it remains unclear whether such strategies can effectively enhance the persistence of γδ T cells within immunocompetent human TMEs. Therefore, identifying cell-intrinsic factors in γδ T cells that can enhance their in vivo persistence is crucial for amplifying their therapeutic potential in cancer immunotherapy.

γδ T-Cell Engagers

Recent research has also focused on engager molecules that enhance the antitumor function of γδ T cells in vivo. However, zoledronate, a traditional engager molecule, has several characteristics that limit its therapeutic efficacy in cancer treatment. First, zoledronate-mediated Vγ9Vδ2 T-cell activation is solely dependent on BTN3A1 among the BTN3A isoforms. Therefore, its therapeutic potential is reduced in tumors that predominantly express BTN3A2 rather than BTN3A1, such as primary AML blasts (Benyamine et al., 2016). To address this limitation, a recent study developed ICT01, a humanized anti-BTN3A antibody that binds to all 3 BTN3A isoforms. This antibody enhanced the cytotoxic potential of adoptively transferred Vγ9Vδ2 T cells in NSG mouse models (De Gassart et al., 2021).

Second, zoledronate has poor tumor specificity. This is critical for its therapeutic effects, as zoledronate must be taken up by tumor cells to induce the antitumor response of Vγ9Vδ2 T cells. To overcome this challenge, a study developed a zoledronate conjugated to an anti–epidermal growth factor receptor (EGFR) antibody. This antibody-conjugated zoledronate increased the antitumor response of Vγ9Vδ2 T cells compared with unconjugated zoledronate against patient-derived CRC organoids in vitro (Benelli et al., 2022).

Additionally, other studies have developed bispecific antibodies as tumor-specific engager molecules for γδ T cells. A bispecific antibody is a complex molecule that combines 2 distinct conventional antibodies or heavy-chain variable domains. Recently, several studies have developed bispecific antibodies that simultaneously target Vγ9Vδ2 T cells and tumor-associated antigens, such as CD1d for chronic lymphocytic leukemia, CD123 for AML, or EGFR for CRC treatment (De Weerdt et al., 2021; Ganesan et al., 2021; King et al., 2023).

γδ TCR-Based Therapies

In addition to γδ T cells, γδ TCRs have recently gained considerable attention due to their ability to recognize cancer hallmarks, like BTN3A (Xin et al., 2024; Yuan et al., 2023). To enhance the tumor-targeting capabilities of conventional αβ T cells, a study developed bispecific molecules composed of ecto-γδ TCR and anti-CD3 single-chain variable fragment (scFv), known as γδ TCR-antibody bispecific molecules (Van Diest et al., 2021). A similar approach involves engineering conventional αβ T cells with defined γδ TCR (TEGs). TEG001, a representative TEG expressing Vγ9Vδ2 TCRs, is currently being explored in a phase I clinical trial (NTR6541) (Straetemans et al., 2018). Preliminary results from the clinical trial showed that TEG001 had longer persistence in the peripheral blood of human patients than Vγ9Vδ2 T-cell therapy, suggesting a potentially long-lasting antitumor effect within the TME (de Witte et al., 2022; Neelapu et al., 2022).

However, TEG001 struggles to replicate the coordinated activation of Vγ9Vδ2 TILs mediated by other receptors within the TME, potentially leading to suboptimal antitumor responses in vivo. To overcome these limitations, recent studies have developed novel TEG001 variants that express costimulatory CARs, such as NKG2D-4-1BB$_{CD28TM}$ or anti-BTN3A1-4-1BB$_{CD28TM}$. These engineered cells demonstrated enhanced cytotoxicity compared with traditional TEG001 (Hernandez-Lopez et al., 2024). Furthermore, another study investigated the potential effects of neural cell adhesion molecule-1 (NCAM1) on TEGs, and found that active motility could influence Vγ9Vδ2 TCR-mediated antitumor responses (Dekkers et al., 2023).

CONCLUSION

γδ T cells have emerged as critical players in antitumor immunity, demonstrating significant prognostic value across various tumors. In particular, the frequency of Vδ1 TILs has been identified as a key prognostic factor, distinct from Vδ2 TILs. These findings have led to an increase in research on the phenotypic differences between Vδ1 and Vδ2 TILs. Vδ1 TILs exhibit activation and exhaustion phenotypes similar to those of conventional αβ TILs, suggesting that their activation is primarily driven by TCR stimulation. In contrast, Vδ2 TILs express lower levels of activation or exhaustion markers, but still display antitumor functions comparable to those of Vδ1 TILs. These results suggest that distinct activation pathways may underlie these phenotypic differences in Vδ2 TILs. However, the precise mechanisms by which antigens or cytokines stimulate γδ TILs remain unclear.

A recent study also highlights the positive correlation between the frequency of Vδ1 TILs and anti-PD-1 therapy responses in melanoma patients, which was not observed in conventional αβ TILs (Davies et al., 2024). These findings suggest that the mechanisms underlying Vδ1 TIL-mediated responses to anti-PD-1 therapy may differ from those of conventional αβ TILs, although the precise pathways remain unresolved. Furthermore, the roles of other immune checkpoints such as CTLA-4, TIM-3, and LAG-3 in regulating human γδ TILs are not yet fully understood. Elucidating these mechanisms could provide valuable insights into developing novel γδ T-cell therapies.

Despite recent advances in γδ T-cell therapies, both Vδ1 and Vδ2 TILs have significant challenges that need to be addressed for successful treatment outcomes. Vδ1 TILs display functional heterogeneity in their immune response to tumors, while Vδ2 TILs exhibit limited proliferative capacity, leading to poor persistence within the TME. If these challenges can be overcome, γδ T-cell therapies have the potential to become a powerful platform in cancer immunotherapy, offering new hope to patients with difficult-to-treat cancers.

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Heung Kyu Lee: Writing—review and editing, writing—original draft, supervision, funding and acquisition. Won Hyung Park: Writing—original draft.

DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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